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ORIGINAL ARTICLE

Agreement between multifrequency BIA and DXA for assessing segmental appendicular skeletal muscle mass in older adults

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Abstract

Background Aging presents changes in muscle mass that may lead to sarcopenia. Identifying safe, quick, and accessible methods to assess muscle mass is imperative.

Aims The purpose of this investigation was to compare the assessments of appendicular skeletal muscle mass (ASMM), fat-free mass (FFM), and fat mass (FM) between bioelectrical impedance analysis (BIA) and dual-energy X-ray absorptiometry (DXA).

Methods Seventy-three healthy, community-dwelling, physically active males ($n = 19$) and females ($n = 54$) (BMI = 27.1 ± 4.3 kg m^{-2}) between the ages of 55–85 years underwent total-body BIA and DXA. ASMM was estimated via BIA from a previously published regression equation while DXA ASMM was calculated as the sum of the measured total arm lean mass and total leg lean mass. Paired-samples t tests with a significance level of $p < 0.05$ were conducted, while agreement between the methods was assessed via Bland–Altman plots.

Results In comparison to DXA, the chosen BIA equation overestimated ASMM (21.61 ± 5.82 kg vs. 18.82 ± 4.81 kg) and FFM (49.57 ± 9.94 kg vs. 46.22 ± 10.11 kg) and underestimated FM (24.59 ± 8.28 kg vs. 27.13 ± 10.01 kg), all $p < 0.001$. Visual inspection of the Bland–Altman plots revealed wide limits of agreement. Female participants were more clustered around the mean than male participants.

Discussion The multifrequency BIA device and chosen ASMM estimation equation resulted in wide limits of agreement and significantly different comparisons to the reference method of DXA.

Conclusion Future research should continue to investigate and validate methodologies to screen older individuals for characteristics of aging-related diseases, such as sarcopenia.

Keywords Fat-free mass · Fat mass · Body composition · Aging · Imaging

Introduction

Bioelectrical impedance analysis (BIA) has emerged as a portable, affordable, and noninvasive modality to assess body composition, skeletal muscle, fat-free, and fat mass. This is of greater importance in older adults, as age-related skeletal muscle atrophy is related to a decreased quality of life and can potentially result in sarcopenia [1, 2]. To attempt the best monitor age-related atrophy, older adults should be screened frequently and early. With screening and detection of any loss, it is then possible for an older adult to engage in lifestyle modifications, such as increased resistance training participation [3] or nutritional interventions [4]. Therefore, it is vital to have valid, noninvasive, easily accessible methods to routinely screen adults in busy clinical settings.

To assess skeletal muscle mass, devices, such as dual-energy X-ray absorptiometry (DXA), BIA, computerized tomography (CT), or magnetic resonance imaging (MRI) have been utilized [5]. From these, measurements, such as segmental or appendicular skeletal muscle mass (ASMM) for the arms and legs can be derived in addition to fat-free mass (FFM) and fat mass (FM) [2]. ASMM is a viable measurement in the assessment of aging adults as it can account for approximately 75% of the total skeletal muscle mass [6, 7] and

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is highly correlated to physical function [8], which correlates to successful performance of activities of daily living.

Although DXA has been classically deemed as the reference method for assessment of ASMM and other body composition variables [6, 9, 10], it is expensive and exposes patients to radiation (although minimal amounts) during the screen and frequent monitoring by this method accumulates the dose acquired by each patient [11]. The only time DXA may be warranted in this population would be to provide additional examination of bone density. DXA devices are relatively uncommon in standard clinical settings as it has predominantly been used in research facilities and specialized clinical facilities. Radiation exposure also occurs with CT [12] and, similar to MRI, these devices are typically not readily available, extremely time consuming, and expensive. Alternatively, BIA is an inexpensive, portable, and noninvasive method for assessing body composition. BIA devices vary and have some variability on accuracy, depending on the condition of the individual it is used on (dehydrated, fatigued, injured, or diseased). Although BIA does not directly measure ASMM, previous works have derived regression equations to estimate ASMM. For example, Janssen et al. identified an accurate estimate of ASMM from BIA using an individual's height, weight, sex, and impedance [2]. Multiple other studies have attempted to expand upon the results of Janssen et al., with conflicting results [6, 10] that might be due to the equation used or the population studied. These other equations, although valid and reliable, use additional data such as resistance-adjusted for height and reactance of a measurement. The more variables included introduce greater error on the variable measurements yet may add to the overall accuracy of the prediction [13]. However, these three potential equations (Janssen et al., Kyle et al., and Sergi et al.), among many others, are all relatively comparable in the present population and could potentially be used interchangeably.

Past studies have attempted to compare the measurements of ASMM, FM, and FFM between DXA and BIA with mixed results based on the ethnicity of the population used [6, 11], BIA frequencies [14], hospitalization status [15], and age [10]. Further, as BIA requires the estimation of ASMM from previously established regression equations, the equation utilized may influence the results, depending on the population [6, 10, 13, 16]. Therefore, the purpose of the present study was to examine the levels of agreement between BIA and DXA to estimate ASMM, FFM, and FM in healthy, midwestern, community-dwelling, recreationally active older adults.

Methods

Study design

This study included participants between the ages of 55 and 85, in good health, as determined by the health and exercise status questionnaire. Measurements of DXA and BIA were performed on all participants in a fasted and rested state. All study procedures were approved for ethical compliance as outlined in the Declaration of Helsinki and approved by the University Institutional Review Board as part of the Human Subjects Protection Program.

Subjects

Healthy adult males and females with no physical limitations were recruited to participate in this investigation. Inclusion criteria involved completion of self-reported health and exercise status including documentation of all physical activity and medications. Exclusion criteria involved device limitations: no nonremovable metal implants or devices, body mass less than 136 kg, or limb amputees. All subjects signed a written statement of informed consent prior to participating in any study-related activities.

Body composition

Participants arrived at the laboratory in a hydrated, fasted, and rested state. They were also asked to remove any excess clothing with zippers or metal clasps as well as to use the restroom prior completing the assessments. Height (cm) and body mass (kg) were assessed using a standard physician scale with an attached stadiometer (Detecto 339, Cardinal Detecto, Webb City, MO, USA). Total body skeletal mass was quantified using dual-energy X-ray absorptiometry (DXA; Lunar Prodigy Advanced, PA + 300532, and Encore software, v.8.8, GE Healthcare, Madison, WI). For the full-body scan, participants were asked to remove any metal jewelry from their body, socks, and shoes and lay supine on the DXA scan table with the hands pronated and within the scan table boundaries. A Velcro strap was secured proximal to the calves and ankles to ensure limited movement during the scan. Segmental body composition values collected included total arm lean mass (TALM; kg) and total leg lean mass (TLLM; kg), total body FFM (kg), and FM (kg). To calculate ASMM, TALM and TLLM were summed.

In addition, participants were asked to lie on their back on a padded examination table for 5–10 min while the investigator cleaned the skin from hair, dirt, or lotion with alcohol for electrode placement. Body tissue impedance was measured with a multifrequency BIA (SFB7; ImpediMed Ltd., Pinkenba, Queensland, Australia). Electrodes were placed on the wrist in line with the lateral epicondyle of the radius and on the ankle between the malleoli. Additional electrodes were placed on the hand and foot, 5 cm from the first electrode's placement. To run the assessment, height, weight, and sex were entered into the device. Impedance values estimated total body water, total body FFM, and total body FM. ASMM mass via BIA was estimated using an equation

established by Janssen et al. in a comparable multiethnic group of adults [2] where Ht is height in cm; R = BIA resistance in Ohms; sex is 0 for females and 1 for males, and age is in years.

$$\text{ASMM (kg)} = \left[\left(\frac{\text{Ht}^2(\text{cm})}{R} \times 0.401\right) + (\text{sex} \times 3.825) + (\text{age} \times -0.071)\right] + 5.102.$$

Statistical analysis

Differences between the two methods were assessed using separate paired sample t tests for ASMM, FFM, and FM. Effect size of the differences were calculated using Cohen's d [17]. In addition, Pearson's correlations were utilized for both methods and all outcome variables. Agreement between the two methods for ASMM, FFM, and FM were assessed by Bland–Altman plots with 95% limits of agreement [18]. Limits of agreement (LOA) were calculated as the mean of the difference between the methods ± 1.96 × the standard deviation of the difference between the methods [18]. Further, intraclass correlation coefficient ($ICC_{2,k}$), minimal clinical difference, and standard error of the measurement were included for all variables [19]. An additional exploratory regression model was constructed using stepwise linear regression to predict ASMM using the SFB7 BIA. Variables initially considered for use in the model included reactance, age, height, body mass, sex, and total body water. All analyses were conducted using Statistical Package for the Social Sciences (SPSS) version 26, (IBM Corp. Armonk, NY, USA) or Excel (Microsoft Office, Redmond, WA, USA) with a predetermined level of significance (α) of 0.05. All data are presented as mean ± SD.

Results

Seventy-three healthy males (n = 19; mean ± SD; age = 69.4 ± 6.3 years; height = 176.6 ± 7.5 cm; body mass = 81.9 ± 15.0 kg) and females (n = 54; age = 66.1 ± 7.2 years; height = 160.9 ± 7.6 cm; body mass = 71.1 ± 12.8 kg) volunteered for this study. The participants would be classified as recreationally active according to the American College of Sports Medicine [20] as they self-reported being engaged in 3.5 ± 2.3 h per week of aerobic activity (primarily walking), 0.5 ± 1.0 h of resistance exercise, and 0.5 ± 1.9 h of sport-related activity (tennis, pickleball, golf, etc.) with no chronic, uncontrolled health conditions or exercise limitations.

The separate paired sample t tests indicated that ASMM ($p < 0.001$), FFM ($p < 0.001$), and FM ($p < 0.001$) were significantly different between the two methods with small to moderate effect sizes. The mean difference ± SEM between BIA and DXA for ASMM was 2.78 ± 1.86 kg (LOA: − 6.58–1.02). While the mean difference for FFM was 3.35 ± 3.52 kg (LOA: − 10.72–4.02) and FM was 2.55 ± 1.64 kg (LOA: − 4.44–9.56). However, the minimal clinical difference for the variables is 3.8 kg, 7.0 kg, and 7.4 kg, suggesting that despite the significant differences between the two methods of assessment for each variable with a t test, the mean differences are less than the minimal difference and the two methods could be considered 'not different'. Table 1 presents average values for all three variables from both methods. Bland–Altman plots of the comparisons for ASMM are shown in Fig. 1a, b displays the FFM comparisons, and Fig. 1c represents FM. Both methods were highly correlated with Pearson's correlation values of $r = 0.95$, $r = 0.93$, and $r = 0.94$ (all $p < 0.001$) for ASMM, FFM, and FM, respectively. Similarly, the ICCs for each respective variable were 0.943, 0.940, and 0.984.

When exploring the possibility of predicting ASMM using the specific 4-electrode SFB7 BIA, the strongest model resulted in a projected equation of:

$$\text{ASMM(kg)} = [(\text{TBW} \times 0.71) + (\text{sex} \times 6.58) + (\text{ht} \times 0.21)] - 18.18,$$

where TBW = total body water; sex (0 = female and 1 = male), and ht = height in cm and were all significant contributors ($p < 0.001$) to the overall model ($R^2 = 0.935$; $p < 0.001$). Further validation of this prediction was performed using an additional 57 males and females of similar characteristics at a later date. A paired t test comparing means of predicted ASMM and actual DXA-derived ASMM resulted in no significant mean difference (0.30 kg; $p = 0.35$).

Table 1 Variable mean ± SD, mean difference 95% confidence intervals (CI), effect size, and correlations between methods

$n = 73$	DXA	BIA	95% CI	Cohen's d	Pearson r
ASMM (kg)	18.82 ± 4.81	21.61 ± 5.82*	(− 3.24, − 2.34)	0.52	0.95*
FFM (kg)	46.22 ± 10.11	49.57 ± 9.94*	(− 4.22, − 2.47)	0.33	0.93*
FM (kg)	27.13 ± 10.01	24.59 ± 8.28*	(1.72, 3.38)	0.27	0.94*

ASMM appendicular skeletal muscle mass, *FFM* fat free mass, *FM* fat mass

*Significant at $p < 0.001$

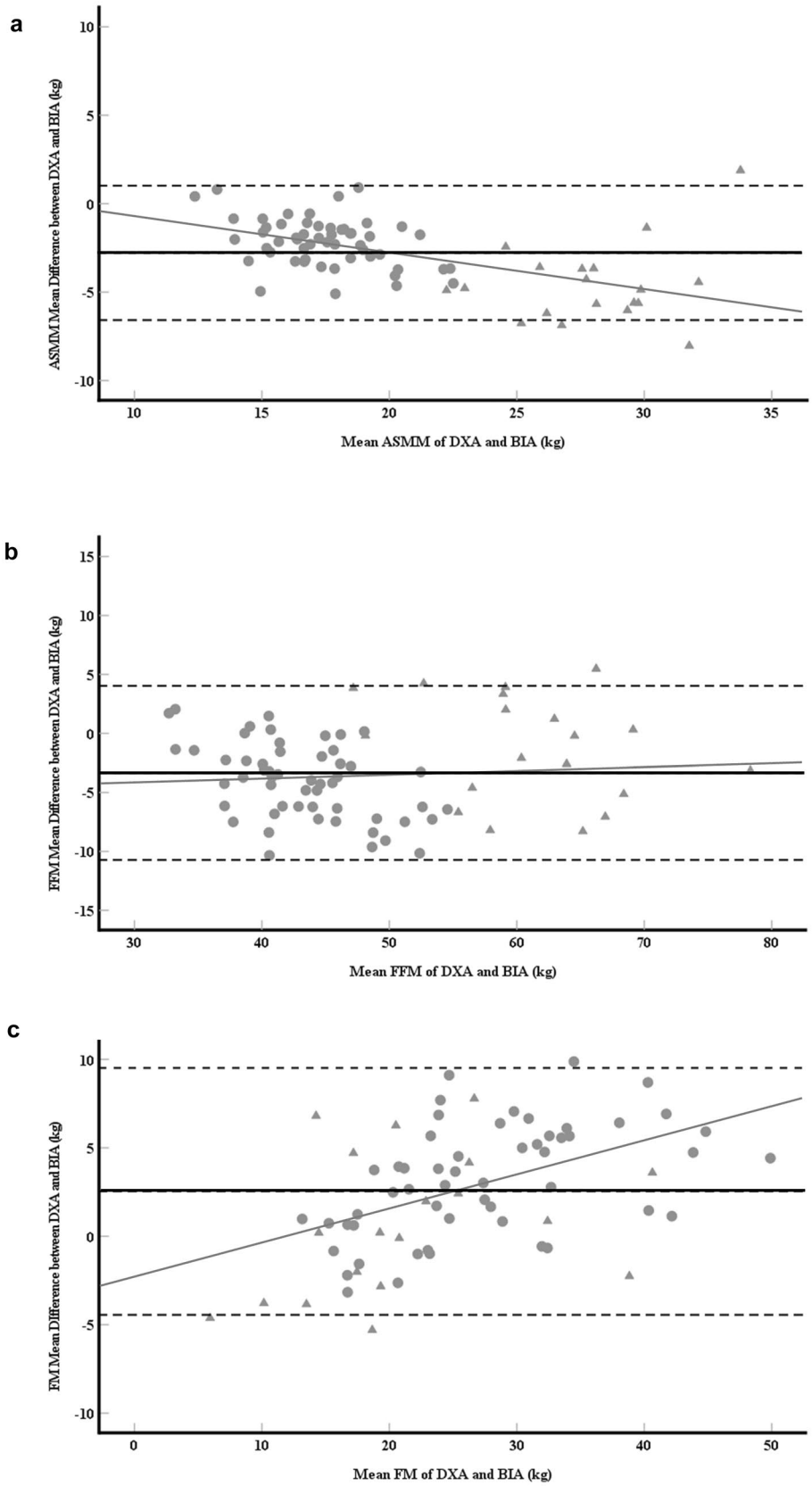

Fig. 1 Bland–Altman plot of the limits of agreement for ASMM (**a**), FFM (**b**), and FM (**c**) between DXA and BIA. Circles = female participants; triangles = male participants; the thick black lines represent the mean; dashed black lines represent the LOA; thick gray line represents the regression trend

Discussion

The purpose of this investigation was to examine the relationship and agreement between BIA and DXA for assessing ASMM, FFM, and FM in older adults. The present results suggest that the multifrequency BIA and estimation equation utilized in this study overestimated ASMM and FFM by approximately 12.9% and 6.7%, respectively, and underestimated FM by approximately 9.4% in these older adults. There was also large dispersion and wide limits of agreements amongst each of the three variables when compared through visual analysis of the Bland–Altman plots.

Despite the many benefits of BIA, such as portability, ease of use, and lack of radiation exposure, the results from the present study suggest that using the Janssen et al. equation for ASMM estimation does not produce similar results of ASMM, FFM, and FM when compared to the reference method of DXA. We have included an additional equation that includes SFB7 BIA-derived TBW, as a reflection of the estimation of hydration within and around cells to predict ASMM. This prediction resulted in no difference from DXA-estimated ASMM in a smaller cohort of similar older adults, however our exploration was not adequately powered for conclusive results and accurate agreement. Thus, this should be further explored with a much larger heterogeneous group to suggest the SFB7-specific equation is valid. Further, although BIA has been utilized extensively in previous research studies [2, 6, 11, 21], its results still vary, and its use can be controversial. This is primarily due to estimation of indirect variables, such as ASMM, rather than a fault of the device itself. A similar methodology and sample to the present study was assessed by Sergi et al., in which 296 Caucasian older adults had ASMM measured via DXA and estimated via BIA. The Kyle et al. [10] and a new equation to estimate ASMM from BIA were utilized, and although both equations were found to have similar results to DXA, the new equation was more reliable [6]. This might have been due to the sample selected, as Kyle et al. measured adults of various ages (20–94 years) [10] and Sergi et al. focused on older adults over 60 years [6], much like the present study. The present study chose to not use the Sergi equation because it was derived in a purely Caucasian cohort and the present sample was multiethnic. Similarly, González Correa et al. evaluated the diagnosis of sarcopenia by DXA and BIA in 72 older adults, 28 of which were diagnosed with sarcopenia via DXA, from Colombia and determined that BIA only correctly diagnosed 57% of the sarcopenic patients. The Janssen et al. equation was utilized by González Correa et al. and resulted in BIA overestimating the skeletal muscle mass index [21], which led to the high misdiagnosis rate. Although sarcopenic status was not assessed in the present study, BIA overestimated ASMM by 2.8 kg when compared to DXA and would likely misdiagnose individuals similar to those in the González Correa study. The European Working Group on Sarcopenia in Older People (EWGSOP) has suggested either method could be used, yet there are different SMI criteria, depending on the use of DXA or BIA in classification of muscle [22]. Further, the LOA of the present study for ASMM were − 6.58–1.02 kg, which is wider than the LOAs reported from Sergi et al. [6], yet similar to a study that utilized the Janssen equation and determined it was not appropriate in older Swiss adults [23]. Therefore, if these methodologies were utilized in a clinical setting with a population at a greater risk for sarcopenia or other aging-related diseases, a diagnosis could have been missed.

Owing to the different results generated and populations studied from previous works, the present study was developed to further investigate the agreement between BIA and DXA in older adults. It is vital to determine if BIA and its associated regression equations to estimate ASMM is a valid screening option when compared to DXA in older adults. Most importantly because sarcopenia can affect up to 34% of adults over 65 years of age [24] and is classified based on the loss of muscle size and strength [1, 25]. In the present study, the Janssen et al. equation was utilized to estimate ASMM, which had been developed from the assessment of 269 Caucasians with an average age of 41.5 years [2]. As individuals age, it is possible that the cross-sectional area of the limbs decreases and adipose tissue is distributed to the trunk to a greater degree [6, 26]. In addition, many older adults have altered hydration status, depending on medication use and various disease states that can impact muscle and cellular composition. Therefore, BIA-based estimations may have greater variability in older adults than other populations. This was confirmed with the more technical ASMM prediction equation developed by Sergi et al.

Similar to ASMM, BIA measurements and respective Janssen equation also overestimated FFM while underestimating FM. We would speculate this may be to the slight difference in age of the evaluated participants. Achamarh et al. determined that participants with a BMI between 25.0 and 30.0 kg m^{-2}, BIA overestimated FFM by 7.1 kg, while underestimating FM by 5.6 kg in subjects with an average age of 33.4 [27]. While the present study sample had an average age of 66.9 years and a BMI of 27.07 kg m^{-2}, similar results were found as the Janssen BIA equation overestimated FFM by only 3.35 kg and underestimated FM by 2.55 kg. The LOA of the present study were for FFM were − 10.72–4.02 and − 4.44–9.56 kg for FM, while Achamarh et al. had LOA of − 15.1–1.1 kg and − 2.0–13.3 kg for FFM and FM, respectively, and determined those LOA were clinically unacceptable [27]. Although this study was not powered to stratify by sex, visual inspection of the

Bland–Altman plots illustrated that there was greater variability among the male participants. Clinically, this may indicate that the Janssen BIA equation resulted in greater overestimation of ASMM in males or subjects with higher ASMM. Therefore, due to significant differences between the means of FFM and FM in the present study and the wide, clinically unacceptable LOA according to Achamarh et al., it can be stated that BIA measurement of FFM and FM in the present cohort of older adults was clinically unacceptable and this may vary by the sex of the participant.

The present study is novel in that it investigated the limits of agreement between BIA and DXA in an older population that was not diagnosed as sarcopenic, community dwelling, and recreationally active. As previous BIA–DXA agreement studies have focused on hospitalized older adults [28], large age ranges [10], and sarcopenic patients [21], this study adds to the knowledge of the ability of BIA to assess body composition related variables in healthy older adults when compared with DXA as the reference method. Future studies should continue to investigate how BIA and DXA compare, and how BIA measurements are influenced by the device utilized (single v. multifrequency), estimation equations, and population characteristics (e.g., age, disease state).

A few limitations of the current study include the relatively homogenous population, cross-sectional design, and characteristics of the participants. Only high functioning subjects, those who could travel to the laboratory were included, which may decrease the generalizability of these results to hospitalized or diseased patients. Also, due to the low number of male subjects (~24% of the total sample), results could not be stratified by sex. However, despite these limitations, this study adds to the knowledge of the ability of BIA to assess skeletal muscle and fat mass related variables when compared with DXA as the reference modality.

From the results of this study, it can be concluded that BIA does not agree with the reference method of DXA for assessing sarcopenic-related body composition variables. The results of this study may have been influenced by the use of a validated equation generated from younger adults. Also, these differences between modalities may differ between sexes. Future studies should continue to investigate various modalities and equations to assist in the assessment of body composition related variables in older adults. However, irrespective of the screening method, it is still imperative for practitioners to continually stress with their aging patients that lifestyle habits, such as sufficient exercise (aerobic and resistance) and diet, are always necessary to help prevent and attenuate the effects of age-related decrements in physical function due to muscle loss.

Author contributions AAH acquired funding for study supplies and obtained IRB approval. AAH collected the data and design the study design. CJC wrote the first draft of the manuscript with AAH's oversight and approval. CJC and AAH reviewed and approved the final version of the manuscript.

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Data availability The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Conflict of interest On behalf of all authors, the corresponding author states that there is no conflict of interest.

Ethical approval All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional research committee and with the 1964 Helsinki Declaration and its later amendments or comparable ethical standards. The study was approved by the Human Research Protection Program at the University of Kansas (No. STUDY00143495).

Informed consent Informed consent was obtained from all individual participants included in the study.

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